Weeks 2–4: Randomized Experiments and Design-Based Causal Inference

Overview and Philosophy

In Weeks 2 through 4, we study causal inference through randomized experimentation. The central message is simple but profound: *randomization itself justifies causal inference.* No assumptions about outcome models, functional forms, or large-sample approximations are required.

This design-based perspective contrasts sharply with regression adjustment studied in Week 1. There, causal interpretation depended on modeling assumptions and implicit covariate balancing. Here, balance is achieved directly through the random assignment mechanism.

Following Neyman, Fisher, and Rosenbaum, we treat potential outcomes as fixed attributes of the experimental units and view probability as arising solely from the randomized assignment.

2 Week 2: Fundamentals of Randomized Experiments

2.1 Randomization as a Source of Covariate Balance

In Week 1, we studied how linear regression achieves covariate adjustment by removing the linear dependence of the treatment variable on the observed covariates. This led to the interpretation of the regression coefficient as a comparison between treated and control units after reweighting (which involves projecting both the outcome and the treatment onto the orthogonal complement of the column space of X). While this perspective is powerful, it also reveals a fundamental limitation: regression balances only those (functions of the) covariates that lie within the linear span of the included regressors. If the true outcome model depends on nonlinear or higher-order functions of X that are not included in the regression, then the balancing property no longer holds, and bias may arise.

Randomization, by contrast, produces a form of covariate balance that is far more extensive. It does not depend on modeling assumptions, functional forms, or linear projections. Instead, balance arises directly from the assignment mechanism itself. This week, we develop both the intuition and the formal mathematics behind this idea, and we show that randomization balances not only the observed covariates but also any function of them, including functions unknown to the researcher. In fact, randomization balances even unobserved covariates, a property that no observational method can guarantee.

Intuition: Randomization as a Shuffling Mechanism

When treatment is assigned by a known random mechanism, the treated and control groups are formed by a process that is independent of all pre-treatment covariates. As a result, any differences between the groups must be due to chance alone, and these differences vanish in expectation. This is the core reason why randomized experiments are the gold standard for causal inference: they create treated and control groups that are comparable in every respect except for the treatment itself.

Setup and Notation

Let X_i denote any pre-treatment covariate for unit i, which may be scalar or vector-valued. Let $Z_i \in \{0, 1\}$ denote the treatment assignment under a completely randomized design with n_1 treated

units and $n_0 = N - n_1$ control units. The assignment mechanism satisfies

$$\Pr(Z = z) = \frac{1}{\binom{N}{n_1}} \qquad \text{for all } z \text{ with } \sum_i z_i = n_1.$$

The defining feature of this design is that Z is statistically independent of X. This single property is responsible for the entire suite of balancing results that follow.

Balance in Means

To begin, consider the treated mean of a scalar covariate X,

$$\bar{X}_T = \frac{1}{n_1} \sum_{i:Z_i=1} X_i.$$

Because Z is independent of X, each unit is equally likely to appear in the treated group, and therefore

$$\begin{aligned} \mathrm{E}[\bar{X}_T] &= \mathrm{E}\left[\frac{1}{n_1}\sum_{i=1}^{N} Z_i X_i\right] \\ &= \frac{1}{n_1}\sum_{i=1}^{N} \mathrm{E}\left[Z_i X_i\right] \\ &= \frac{1}{N}\sum_{i=1}^{N} \mathrm{E}(X_i\ ) = \mathrm{E}(\bar{X}). \end{aligned}$$

An identical argument shows that $\mathrm{E}[\bar{X}_C] = \mathrm{E}(\bar{X})$, so that

$$\mathrm{E}[\bar{X}_T - \bar{X}_C] = 0.$$

This is the simplest form of covariate balance: the treated and control groups have the same expected mean for every covariate. Importantly, this holds for *every* covariate, whether or not the researcher has measured it.

Balance Beyond Means

The balancing property of randomization is not limited to means. For example, the variance of the treated mean is

$$\mathrm{Var}(\bar{X}_T) = \frac{1}{n_1}\left(\frac{N - n_1}{N - 1}\right) S_X^2,$$

where S_X^2 is the finite-population variance of X. A similar expression holds for the control group. Consequently, the difference in means satisfies

$$\mathrm{Var}(\bar{X}_T - \bar{X}_C) = S_X^2 \left(\frac{1}{n_1} + \frac{1}{n_0}\right)\left(\frac{N}{N-1}\right).$$

This shows that the difference in means is not only centered at zero but also becomes more concentrated around zero as the sample size grows. In large experiments, substantial imbalances are exceedingly unlikely.

Balance for Arbitrary Functions of Covariates

A striking feature of randomization is that it balances not only the covariates themselves but also *any* function of them. Let $g : \mathbb{R}^p \to \mathbb{R}$ be any function, possibly nonlinear, discontinuous, or unknown to the researcher. Define

$$\bar{g}_T = \frac{1}{n_1} \sum_{i:Z_i=1} g(X_i), \qquad \bar{g}_C = \frac{1}{n_0} \sum_{i:Z_i=0} g(X_i).$$

Because Z is independent of X, it follows immediately that

$$\mathrm{E}[\bar{g}_T] = \mathrm{E}[\bar{g}_C] = \frac{1}{N} \sum_{i=1}^{N} g(X_i),$$

and hence

$$\mathrm{E}[\bar{g}_T - \bar{g}_C] = 0.$$

This result holds for every function g, including functions the researcher does not know or cannot measure. In particular, randomization balances all nonlinear transformations, all interactions, and all higher-order moments of the covariates. This is in stark contrast to regression, which balances only those functions of X that lie in the linear span of the included regressors.

Distributional Balance

The balancing property of randomization extends even further. Let $F_T(x)$ and $F_C(x)$ denote the empirical distribution functions of a covariate X in the treated and control groups. Under randomization,

$$\mathrm{E}[F_T(x)] = \mathrm{E}[F_C(x)] = F(x),$$

the population distribution function. Thus, randomization balances not only means and variances but also entire distributions. For any measurable set A,

$$\mathrm{E}\left[\frac{1}{n_1} \sum_{i:Z_i=1} \mathbf{1}\{X_i \in A\}\right] = \mathrm{E}\left[\frac{1}{n_0} \sum_{i:Z_i=0} \mathbf{1}\{X_i \in A\}\right].$$

This is a form of distributional balance that no regression adjustment can replicate.

Balance for Unobserved Covariates

Perhaps the most important implication of randomization is that it balances even unobserved covariates. Let U_i denote any unmeasured pre-treatment characteristic, such as genetic predisposition, motivation, or personality traits. Even though the researcher does not observe U_i, the independence of Z and U implies

$$\mathrm{E}[\bar{U}_T - \bar{U}_C] = 0.$$

This is the fundamental reason why randomized experiments eliminate confounding: they ensure comparability between treated and control groups along all pre-treatment dimensions, including those that are unmeasured or even unknown.

Summary

Randomization produces a level of covariate balance that far exceeds what can be achieved by regression or other observational methods. It balances means, variances, and entire distributions; it balances all functions of the covariates, whether known or unknown; and it balances both observed and unobserved covariates. These properties arise directly from the assignment mechanism and do not depend on any modeling assumptions. This is why randomized experiments are the gold standard for causal inference: they create treated and control groups that are comparable in every respect except for the treatment itself, thereby providing a firm foundation for credible causal conclusions.

2.2 *More Generally,* What Is a Randomized Experiment?

A randomized experiment is defined by its *assignment mechanism*, not by its outcomes. The investigator specifies a probability distribution over treatment assignments and then carries out one realization of this random process.

Formally, we observe/select a finite set of N units, indexed by $i = 1, \ldots, N$, and assign each unit to treatment ($Z_i = 1$) or control ($Z_i = 0$). The defining feature is that **the joint distribution of the assignment vector**

$$\mathbf{Z} = (Z_1, \ldots, Z_N)$$

is known and controlled.

This knowledge is what allows us to make exact probability statements about estimators and test statistics.

Potential Outcomes and the Causal Question

For each unit i, define two potential outcomes:

$$Y_i(1) \quad \text{and} \quad Y_i(0),$$

representing the outcomes that would be observed under treatment and control, respectively.

Consequently, we observe $Y_i = Y_i(1)$ when $Z_i = 1$ and $Y_i = Y_i(0)$ when $Z_i = 0$.

The individual causal effect,

$$\tau_i = Y_i(1) - Y_i(0),$$

is never observable for any unit. This is the *fundamental problem of causal inference*: causal effects are defined as comparisons between potential outcomes that cannot be simultaneously observed.

Estimand: Finite-Population Estimand

In randomized experiments, we do not assume that units are sampled from a population. Instead, we condition on the observed units and their (fixed but unknown) potential outcomes.

The primary estimand is the finite-population average treatment effect:

$$\tau = \frac{1}{N} \sum_{i=1}^{N} \left[Y_i(1) - Y_i(0) \right].$$

This quantity is deterministic. Probability enters only when we consider how treatment is randomly assigned.

The Assignment Mechanism

Let $\mathbf{Z}$ denote the assignment vector. The assignment mechanism is the probability distribution $\Pr(\mathbf{Z} = \mathbf{z})$. As an illustration we consider the following simple, widely studied assignment mechanism.

Completely Randomized Design. Fix integers n_1 and $n_0 = N - n_1$. All assignments satisfying $\sum_i Z_i = n_1$ are equally likely:

$$\Pr(\mathbf{Z} = \mathbf{z}) = \begin{cases} \binom{N}{n_1}^{-1}, & \sum_i z_i = n_1, \\ 0, & \text{otherwise.} \end{cases}$$

Each unit has probability n_1/N of receiving treatment, but assignments are not independent across units.

Estimator: Difference in Means

The most natural estimator of τ compares the average outcomes in the treated and control groups:

$$\hat{\tau} = \bar{Y}_1 - \bar{Y}_0 := \frac{1}{n_1} \sum_{i:Z_i=1} Y_i - \frac{1}{n_0} \sum_{i:Z_i=0} Y_i.$$

This estimator exploits the symmetry induced by randomization: treated and control groups differ only because of the random assignment.

3 Neymanian Inference in Randomized Experiments

One prevailing philosophy of statistical inference in a randomized experiment is the known assignment mechanism. **This perspective, developed by Neyman (1923) and later refined by Fisher and Rosenbaum, treats potential outcomes as fixed attributes of the units and views probability as arising solely from the randomization.** In this section, we develop Neyman's framework in detail, beginning with the estimand, then the estimator, and finally the variance of the estimator under the randomization distribution. We conclude by contrasting this design-based approach with the classical two-sample t-test, which relies on distributional assumptions about outcomes rather than the design of the experiment.

3.1 The Finite-Population Estimand

Consider N units, indexed by $i = 1, \ldots, N$. For each unit, define two potential outcomes:

$$Y_i(1), \qquad Y_i(0),$$

representing the outcomes that would be observed under treatment and control, respectively. These potential outcomes are treated as fixed, unknown quantities. The only randomness in the problem comes from the assignment vector $Z = (Z_1, \ldots, Z_N)$, where $Z_i = 1$ if unit i is assigned to treatment and $Z_i = 0$ otherwise.

The primary estimand in Neyman's framework is the *finite-population average treatment effect* (PATE):

$$PATE = \tau = \frac{1}{N} \sum_{i=1}^{N} \big(Y_i(1) - Y_i(0)\big).$$

This quantity is not a parameter of a probability distribution; it is a fixed number describing the difference in potential outcomes for the units actually in the study.

3.2 Observed Outcomes and the Estimator

The observed outcome for unit i is

$$\boxed{Y_i^{\text{obs}} = Z_i Y_i(1) + (1 - Z_i) Y_i(0).}$$

Under a completely randomized design with n_1 treated units and $n_0 = N - n_1$ control units, the natural estimator of τ is the difference in sample means:

$$\hat{\tau} = \bar{Y}_T - \bar{Y}_C = \frac{1}{n_1} \sum_{i:Z_i=1} Y_i^{\text{obs}} - \frac{1}{n_0} \sum_{i:Z_i=0} Y_i^{\text{obs}}.$$

This estimator exploits the symmetry induced by randomization: treated and control groups differ only because of the random assignment.

For a different design (i.e., cluster randomized design) the estimator may be different.

3.3 Unbiasedness Under Randomization

Because treatment assignment is independent of the potential outcomes, each unit has probability n_1/N of being assigned to treatment. Therefore,

$$\mathrm{E}[\bar{Y}_T] = \frac{1}{n_1} \sum_{i=1}^{N} \mathrm{E}[Z_i] Y_i(1) = \frac{1}{N} \sum_{i=1}^{N} Y_i(1),$$

and similarly,

$$\mathrm{E}[\bar{Y}_C] = \frac{1}{N} \sum_{i=1}^{N} Y_i(0).$$

Subtracting yields

$$\mathrm{E}[\hat{\tau}] = \tau.$$

Thus, the difference-in-means estimator is unbiased for the PATE under the completely randomized design. No modeling assumptions or distributional assumptions are required; unbiasedness follows directly from the assignment mechanism.

3.4 Neyman's Exact Variance Formula

To derive the variance of $\hat{\tau}$, define the finite-population variances

$$S_1^2 = \frac{1}{N-1} \sum_{i=1}^{N} \big(Y_i(1) - \bar{Y}(1)\big)^2, \qquad S_0^2 = \frac{1}{N-1} \sum_{i=1}^{N} \big(Y_i(0) - \bar{Y}(0)\big)^2,$$

and the finite-population variance of individual treatment effects,

$$S_\tau^2 = \frac{1}{N-1} \sum_{i=1}^{N} \big((Y_i(1) - Y_i(0)) - \tau\big)^2.$$

A standard randomization argument yields the exact variance:

$$\boxed{\mathrm{Var}(\hat{\tau}) = \frac{S_1^2}{n_1} + \frac{S_0^2}{n_0} - \frac{S_\tau^2}{N}.}$$

The final term involves the joint distribution of $Y_i(1)$ and $Y_i(0)$ and is therefore unobservable.

The proof is in the appendix.

3.5 Conservative Variance Bounds and a Sharper Inequality

The difficulty is that S_τ^2 is not identifiable, because we never observe both $Y_i(1)$ and $Y_i(0)$ for the same unit. To proceed, we bound this term.

The usual conservative bound

The simplest observation is that $S_\tau^2 \geq 0$. Plugging this into Neyman's formula yields

$$\mathrm{Var}(\hat{\tau}) = \frac{S_1^2}{n_1} + \frac{S_0^2}{n_0} - \frac{S_\tau^2}{N} \;\leq\; \frac{S_1^2}{n_1} + \frac{S_0^2}{n_0}.$$

Thus, we get the conservative variance bound of the Neyman's estimator is

$$\boxed{\mathrm{Var}(\hat{\tau}) \leq \frac{S_1^2}{n_1} + \frac{S_0^2}{n_0}.}$$

3.6 Estimating the Variance

Thus, this gives the familiar conservative bound and motivates the standard Neyman variance estimator Since S_1^2 and S_0^2 are finite-population quantities, we estimate them using the sample variances of the observed treated and control outcomes:

$$s_1^2 = \frac{1}{n_1 - 1} \sum_{i:Z_i=1} \left(Y_i^{\mathrm{obs}} - \bar{Y}_T\right)^2, \qquad s_0^2 = \frac{1}{n_0 - 1} \sum_{i:Z_i=0} \left(Y_i^{\mathrm{obs}} - \bar{Y}_C\right)^2.$$

Neyman's first conservative variance estimator is then

$$\widehat{\mathrm{Var}}(\hat{\tau}) = \frac{s_1^2}{n_1} + \frac{s_0^2}{n_0}.$$

This estimator is conservative because it ignores the negative term S_τ^2/N in the true variance.

Constant treatment effects and exactness

Under constant treatment effects, we have

$$Y_i(1) = Y_i(0) + \tau \quad \text{for all } i.$$

In this case, the individual treatment effects do not vary, so

$$S_\tau^2 = 0.$$

Moreover, the variances of the potential outcomes are equal:

$$S_1^2 = S_0^2 =: S^2.$$

Substituting into Neyman's exact variance formula gives

$$\mathrm{Var}(\hat{\tau}) = \frac{S^2}{n_1} + \frac{S^2}{n_0} = S^2 \left(\frac{1}{n_1} + \frac{1}{n_0} \right).$$

The usual conservative bound

$$\frac{S_1^2}{n_1} + \frac{S_0^2}{n_0} = S^2 \left(\frac{1}{n_1} + \frac{1}{n_0} \right)$$

is therefore *exact* when treatment effects are constant.

3.7 A Second formula of the variance

Define the finite-population covariance

$$S_{10} = \frac{1}{N-1} \sum_{i=1}^{N} \big(Y_i(1) - \bar{Y}(1)\big)\big(Y_i(0) - \bar{Y}(0)\big).$$

Then a straightforward algebraic calculation shows that

$$S_\tau^2 = S_1^2 + S_0^2 - 2S_{10}.$$

To see this, write the individual treatment effect as

$$Y_i(1) - Y_i(0) - \tau = \big(Y_i(1) - \bar{Y}(1)\big) - \big(Y_i(0) - \bar{Y}(0)\big),$$

and expand the square:

$$\begin{aligned} S_\tau^2 &= \frac{1}{N-1} \sum_{i=1}^{N} \big((Y_i(1) - \bar{Y}(1)) - (Y_i(0) - \bar{Y}(0))\big)^2 \\ &= \frac{1}{N-1} \sum_{i=1}^{N} \Big[(Y_i(1) - \bar{Y}(1))^2 + (Y_i(0) - \bar{Y}(0))^2 - 2(Y_i(1) - \bar{Y}(1))(Y_i(0) - \bar{Y}(0)) \Big] \\ &= S_1^2 + S_0^2 - 2S_{10}. \end{aligned}$$

The exact variance formula can be rewritten using the identity

$$S_\tau^2 = S_1^2 + S_0^2 - 2S_{10},$$

where S_{10} is the finite-population covariance between $Y_i(1)$ and $Y_i(0)$.

Plugging this equality into the Neyman exact variance formula we get

$$\mathrm{Var}(\hat{\tau}) = \frac{S_1^2}{n_1} + \frac{S_0^2}{n_0} - \frac{S_1^2 + S_0^2 - 2S_{10}}{N} \tag{6}$$

$$= \frac{n_0}{n_1 N} S_1^2 + \frac{n_1}{n_0 N} S_0^2 + 2\frac{S_{10}}{N} \tag{7}$$

3.8 Contrast with the Classical Two-Sample t-Test

The classical two-sample t-test arises from a very different set of assumptions. In the t-test framework, the observed outcomes in the treated and control groups are assumed to be independent random samples from two populations with means μ_1 and μ_0 and variances σ_1^2 and σ_0^2. The estimand is the difference in population means, $\mu_1 - \mu_0$, and the variance of the difference-in-means estimator is derived from distributional assumptions about the outcomes.

In contrast, Neyman's framework does not assume that outcomes are random draws from any population. The potential outcomes are fixed, and the only randomness comes from the assignment mechanism. The estimand is the finite-population ATE, not a population parameter. The variance of $\hat{\tau}$ is derived from the randomization distribution, not from a model for Y.

The classical t-test typically uses the pooled variance estimator

$$s_p^2 = \frac{(n_1 - 1)s_1^2 + (n_0 - 1)s_0^2}{n_1 + n_0 - 2},$$

and the standard error of $\hat{\tau}$ is

$$\mathrm{SE}_t(\hat{\tau}) = s_p \sqrt{\frac{1}{n_1} + \frac{1}{n_0}}.$$

This estimator is unbiased only under the assumption that the two populations have equal variances. Moreover, the validity of the t-test relies on normality assumptions or large-sample approximations.

Neyman's variance estimator,

$$\widehat{\mathrm{Var}}(\hat{\tau}) = \frac{s_1^2}{n_1} + \frac{s_0^2}{n_0},$$

does not pool variances and does not require equal variances or normality.

3.9 Asymptotic Normality and Confidence Intervals

So far, we have derived the expectation and variance of the difference-in-means estimator $\hat{\tau}$ under Neyman's finite-population framework. These results are purely design-based: they rely only on the randomization mechanism and do not assume anything about the distribution of the potential outcomes. However, to construct approximate confidence intervals for τ, we also need a distributional approximation for $\hat{\tau}$ itself. In practice, this is provided by a central limit theorem for randomized experiments.

Heuristic normal approximation

Intuitively, $\hat{\tau}$ is a difference of two sample means, each formed by averaging many terms. Under mild regularity conditions (for example, that the potential outcomes have bounded second moments and that both n_1 and n_0 grow with N), the randomization distribution of $\hat{\tau}$ is approximately normal in large samples. Formally, one can show that

$$\frac{\hat{\tau} - \tau}{\sqrt{\mathrm{Var}(\hat{\tau})}} \xrightarrow{d} \mathcal{N}(0,1) \quad \text{as } N \to \infty,$$

where the variance is the Neyman variance derived earlier. This is a central limit theorem in the finite-population, randomization-based setting: the potential outcomes are fixed, and the only source of randomness is the assignment vector Z.

In practice, we do not know $\mathrm{Var}(\hat{\tau})$, so we replace it with the conservative estimator

$$\widehat{\mathrm{Var}}(\hat{\tau}) = \frac{s_1^2}{n_1} + \frac{s_0^2}{n_0},$$

where s_1^2 and s_0^2 are the sample variances of the observed treated and control outcomes. Under the same regularity conditions, this estimator is consistent for the upper bound on $\mathrm{Var}(\hat{\tau})$, and the standardized statistic

$$\frac{\hat{\tau} - \tau}{\sqrt{\widehat{\mathrm{Var}}(\hat{\tau})}}$$

is approximately standard normal in large samples.

A note on the Central Limit Theorem. It is important to emphasize that the asymptotic normality used to justify Neymanian confidence intervals does *not* come from the classical i.i.d. Central Limit Theorem. In Neyman's framework, the potential outcomes are treated as fixed, non-random quantities. The only randomness comes from the random assignment of units to treatment and control. Thus, the sampling distribution of $\hat{\tau}$ is a consequence of the *randomization mechanism*, not of any probabilistic model for the outcomes.

The appropriate tool is therefore a *finite-population* Central Limit Theorem, which establishes that the difference-in-means estimator is asymptotically normal under repeated randomization, provided that no single unit dominates the finite population and that both treatment and control group sizes grow proportionally with N. A formal statement of this result is provided in Appendix 3.12.

Approximate confidence intervals

The asymptotic normality of $\hat{\tau}$ justifies the usual Wald-type confidence intervals in the Neyman framework. For a nominal level $1-\alpha$ (for example, 95%), an approximate $(1-\alpha)$ confidence interval for τ is

$$\hat{\tau} \;\pm\; z_{1-\alpha/2}\sqrt{\widehat{\mathrm{Var}}(\hat{\tau})},$$

where $z_{1-\alpha/2}$ is the $(1-\alpha/2)$ quantile of the standard normal distribution (for example, $z_{0.975} \approx 1.96$ for a 95% interval).

It is important to emphasize what underlies this interval. The center $\hat{\tau}$ is unbiased for the finite-population ATE τ under the completely randomized design. The variance estimator $\widehat{\mathrm{Var}}(\hat{\tau})$ is conservative for the true randomization variance. The normal quantile $z_{1-\alpha/2}$ comes from an asymptotic approximation to the randomization distribution of $\hat{\tau}$, not from any assumption that the outcomes themselves are normally distributed.

Comparison with the classical t-test

This perspective clarifies the difference between Neymanian inference and the classical two-sample t-test. In the t-test framework, normality (or large-sample approximations) is assumed for the outcome distributions, and the variance estimator is derived from a model in which outcomes are independent draws from two populations.

> *In Neyman's framework, by contrast, the potential outcomes are fixed, and the only randomness comes from the assignment. The approximate normality of $\hat{\tau}$ is a consequence of a central limit theorem for the randomization distribution, and the variance estimator is justified by the design.*

In summary, Neyman's approach provides:

- an unbiased estimator $\hat{\tau}$ of the finite-population ATE,
- a conservative variance estimator $\widehat{\mathrm{Var}}(\hat{\tau})$ based solely on the design,
- and, via a randomization-based central limit theorem, an approximate normal distribution for $\hat{\tau}$ that justifies Wald-type confidence intervals.

These ingredients together yield a coherent, design-based theory of estimation and uncertainty quantification in randomized experiments.

APPENDIX

3.10 High-Probability Balance

Beyond expectations, randomization also provides high-probability guarantees. For example, Hoeffding's inequality implies that for any bounded covariate X,

$$\Pr\left(|\bar{X}_T - \bar{X}_C| > \epsilon\right) \leq 2\exp\left(-\frac{2n_1 n_0}{N}\frac{\epsilon^2}{(b-a)^2}\right),$$

where $X \in [a, b]$. This shows that large imbalances are exponentially unlikely, and the probability of imbalance decreases rapidly as the sample size increases. Similar concentration results hold for vector-valued covariates and for empirical distribution functions.

3.11 Variance of the Difference-in-Means Estimator

We now derive the variance of $\hat{\tau}$ under Neyman's repeated-sampling framework. Throughout this derivation, potential outcomes are treated as fixed, and randomness arises solely from the assignment vector $\mathbf{Z}$.

Recall the estimator:

$$\hat{\tau} = \frac{1}{n_1}\sum_{i=1}^{N} Z_i Y_i(1) - \frac{1}{n_0}\sum_{i=1}^{N}(1 - Z_i)Y_i(0).$$

Define the finite-population means:

$$\bar{Y}(1) = \frac{1}{N}\sum_{i=1}^{N} Y_i(1), \qquad \bar{Y}(0) = \frac{1}{N}\sum_{i=1}^{N} Y_i(0).$$

Define centered potential outcomes:

$$\tilde{Y}_i(1) = Y_i(1) - \bar{Y}(1), \qquad \tilde{Y}_i(0) = Y_i(0) - \bar{Y}(0).$$

Then we may rewrite

$$\hat{\tau} - \tau = \frac{1}{n_1}\sum_{i=1}^{N} Z_i \tilde{Y}_i(1) - \frac{1}{n_0}\sum_{i=1}^{N}(1 - Z_i)\tilde{Y}_i(0).$$

Key Randomization Moments

Under complete randomization:

$$\mathbb{E}(Z_i) = \frac{n_1}{N}, \qquad \mathrm{Var}(Z_i) = \frac{n_1 n_0}{N^2},$$
$$\mathrm{Cov}(Z_i, Z_j) = -\frac{n_1 n_0}{N^2(N-1)}, \quad i \neq j.$$

These covariances reflect the fact that assignments are not independent.

Variance Calculation

Using bilinearity of variance and covariance,

$$\mathrm{Var}(\hat{\tau}) = \mathrm{Var}\left(\frac{1}{n_1}\sum_i Z_i \tilde{Y}_i(1)\right) + \mathrm{Var}\left(\frac{1}{n_0}\sum_i (1-Z_i)\tilde{Y}_i(0)\right) - 2\,\mathrm{Cov}(\cdot)\,.$$

After expanding terms and simplifying using the covariance structure of $\mathbf{Z}$, one obtains

$$\mathrm{Var}(\hat{\tau}) = \frac{S_1^2}{n_1} + \frac{S_0^2}{n_0} - \frac{S_\tau^2}{N},$$

where

$$S_1^2 = \frac{1}{N-1}\sum_{i=1}^N \tilde{Y}_i(1)^2, \quad S_0^2 = \frac{1}{N-1}\sum_{i=1}^N \tilde{Y}_i(0)^2,$$
$$S_\tau^2 = \frac{1}{N-1}\sum_{i=1}^N (\tau_i - \tau)^2\,.$$

Interpretation. The variance depends not only on outcome variability under treatment and control, but also on heterogeneity in individual treatment effects. This third component is unidentifiable because it involves both potential outcomes for the same unit.

3.12 Finite-Population Central Limit Theorem

This appendix states a version of the finite-population Central Limit Theorem (CLT) appropriate for Neymanian randomization inference. The result applies to sampling without replacement from a finite population and underlies the asymptotic normality of the difference-in-means estimator in randomized experiments.

Setup

Let $\{x_{1,N}, x_{2,N}, \ldots, x_{N,N}\}$ be a sequence of finite populations indexed by N. For each N, define the finite-population mean and variance

$$\bar{x}_N = \frac{1}{N}\sum_{i=1}^N x_{i,N}, \qquad S_N^2 = \frac{1}{N-1}\sum_{i=1}^N (x_{i,N} - \bar{x}_N)^2.$$

Suppose we draw a simple random sample without replacement of size n_N from this population, where

$$\frac{n_N}{N} \longrightarrow \rho \in (0,1) \qquad \text{as } N \to \infty.$$

Let $\bar{X}_{n_N}$ denote the sample mean.

Regularity condition

Assume the Lindeberg-type condition

$$\max_{1\le i\le N} \frac{(x_{i,N}-\bar{x}_N)^2}{NS_N^2} \longrightarrow 0. \tag{A1}$$

This condition rules out the possibility that a single unit contributes a non-negligible fraction of the total variance.

Finite-Population CLT

Theorem 3 (Finite-Population Central Limit Theorem)**.** *Under the setup above, if the regularity condition* (A1) *holds, then*

$$\frac{\bar{X}_{n_N}-\bar{x}_N}{\sqrt{(1-\rho)\,S_N^2/n_N}} \xrightarrow{d} \mathcal{N}(0,1) \qquad \textit{as } N\to\infty.$$

Application to Randomized Experiments

To apply this theorem to randomized experiments, define the finite population

$$x_{i,N} = Y_i(1) - Y_i(0),$$

or more commonly, apply the theorem separately to the treated and control potential outcomes. Under complete randomization with n_1 treated units and n_0 control units satisfying

$$\frac{n_1}{N} \to \rho_1 \in (0,1), \qquad \frac{n_0}{N} \to \rho_0 = 1-\rho_1,$$

the difference-in-means estimator can be written as

$$\hat{\tau} - \tau = (\bar{Y}_T - \bar{Y}(1)) - (\bar{Y}_C - \bar{Y}(0)),$$

a difference of two sample means drawn without replacement from the finite populations $\{Y_i(1)\}$ and $\{Y_i(0)\}$.

Applying Theorem 3 to each term and using independence induced by the randomization, we obtain

$$\frac{\hat{\tau}-\tau}{\sqrt{\mathrm{Var}(\hat{\tau})}} \xrightarrow{d} \mathcal{N}(0,1),$$

where $\mathrm{Var}(\hat{\tau})$ is Neyman's exact variance

$$\mathrm{Var}(\hat{\tau}) = \frac{S_1^2}{n_1} + \frac{S_0^2}{n_0} - \frac{S_\tau^2}{N}.$$

Thus, the asymptotic normality of $\hat{\tau}$ arises from the randomization mechanism and the finite-population CLT, not from any assumption that the outcomes are i.i.d. or normally distributed.

Homeworks

Problem 3 (Blocked and Stratified Randomization)**.** *Consider an experiment with N units partitioned into K blocks $\mathcal{B}_1, \ldots, \mathcal{B}_K$, with $|\mathcal{B}_k| = n_k$.*

1. *Define formally the assignment mechanism for a blocked randomized experiment in which m_k units are treated in block k.*
2. *Show that this assignment mechanism reduces to complete randomization when $K = 1$.*
3. *Explain why blocking can reduce variance of the difference-in-means estimator when potential outcomes vary systematically across blocks.*

Problem 4 (Other Randomized Assignment Models)**.** *In addition to complete randomization, consider the following assignment mechanisms:*

- *Bernoulli trials with* $\Pr(Z_i = 1) = p$
- *Cluster randomization*
- *Paired randomization*

For each mechanism:

1. *Describe the assignment rule mathematically.*
2. *State whether the total number of treated units is fixed or random.*
3. *Discuss one advantage and one disadvantage relative to complete randomization.*